Physics Test - 2021

1) (i) Vector quantities are quantities that require both magnitude and direction. For example, velocity where the speed of the object is given as well as the direction in which its travelling.

(ii) One Dimensional motion is motion in a straight line.

2)

	Distance	Displacement
1-	Is a scalar quantity	Is a vector quantity
2-	~~Ea~~ It depends on the path followed by the object.	~~It is not dependent on~~ Its not dependent on the path followed by the object.

3) (i) $\text{speed} = \frac{d}{t} = \frac{100\,m}{2\,s} = 50\ m/s$

(ii)

$$t = \frac{d}{s} = \frac{\cancel{1000}^{20}\,\cancel{m}}{50\,\cancel{m}/s} = 20\,s$$

4) When a train travels from A to B and returns back from B to A, the ~~distance~~ train would have an avg speed of 2 times the distance between A and B / the total time taken.

But the avg velocity of the train would be 0 because there hasn't been any displacement.

5) (i) $u = 2\,m/s$ $\quad a = \frac{(v-u)}{t}$

$v = 8\,m/s$ $\quad \therefore$ ~~speed~~ acceleration

$t = 12\,s$ $\quad \Rightarrow \frac{8-2}{12}$ $\quad = 0.5\,m/s^2$

$\Rightarrow \frac{6}{12}\ \frac{1}{2}$

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iii)

$S = ut + \frac{1}{2}at^2$

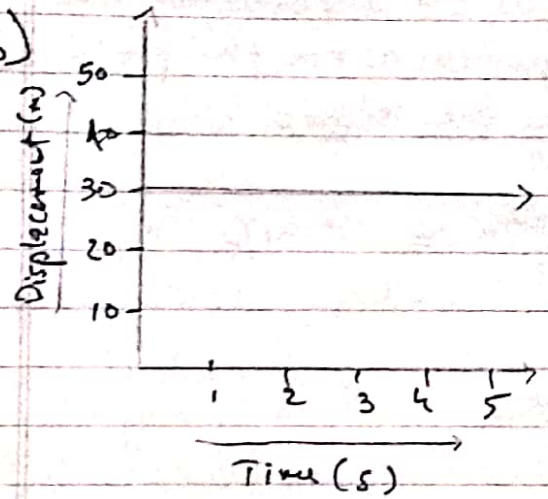
Area of the triangle $= \frac{1}{2} \times (8-2) \times 12 + 2 \times 12$

$\Rightarrow \left(\frac{1}{2} \times 6 \times 12\right) + 24$

$\Rightarrow 36 + 24$

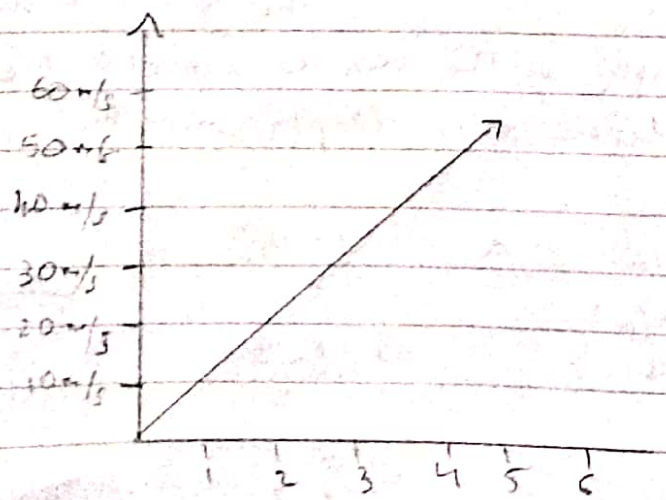
$\Rightarrow 60m$

6)



7) The slope of a displacement-time graph indicates the velocity of the object.

8)



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9)

$$\boxed{S = ut + \frac{1}{2}at^2}$$

<u>derivation</u> :-

$\Rightarrow S = u \times t + \frac{1}{2} \times (v-u) \times t$

$\Rightarrow S = ut + \frac{1}{2} \times at \times t$

$\Rightarrow S = ut + \frac{1}{2}at^2$

10) (i) A has a greater acceleration

(ii) ~~Because the slope of object~~

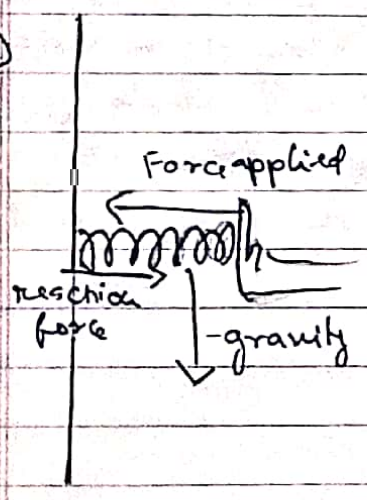
~~Because A covered up with the 13 cm~~

Because the slope of A is more than 13

11) (i) Friction

(ii) Magnetic force

12)



13) Both of them are units of Force

14) $F = ma$

$\Rightarrow 0.5 \times 5$

$\Rightarrow 2.5\ N$

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15) (i) To every action, there is always an equal and opposite reaction.

(ii) For example when someone punches a wall, they apply an action force. Here the wall also applies the same magnitude of force on the fist but in the opposite direction. which causes the pain in our fist after punching the wall

16)

(i) $F = G \times \frac{m_1 \times m_2}{r^2}$

(ii) The force would decrease by 4 times the origin force

17) Because when the glassware falls onto the matress, the time taken to bring it to rest is more which reduces the Force that will be applied on the glassware. ~~F~~ 1/t because $F \propto t$. Therefore it does not break.

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